

Convergence of explicit one-step methods: a short proof and a numerical illustration

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Abstract. We give a compact, self-contained proof that an explicit one-step method of consistency order p converges with global order p for Lipschitz initial value problems. The bound is then compared with numerical experiments for the Euler, Heun and classical Runge–Kutta methods on a linear test equation. The observed orders match the theory, and a cost comparison shows why higher-order methods are preferred at tight tolerances.

Keywords: initial value problems, Runge–Kutta methods, global error, order of convergence

1 INTRODUCTION

Explicit one-step methods for ordinary differential equations go back to Euler [3] and were extended by Runge, Heun and Kutta [5, 7, 9]. Their convergence theory is classical and is treated in many textbooks [1, 2, 4, 10].

This note has two goals. First, to present the global error estimate in a form that is short enough to be proved in a few lines (Section 3). Second, to check the estimate against computed errors for three standard methods (Section 4). All figures, tables and the algorithm are typeset in L^AT_EX [6, 8], and the data behind the plots are computed, not measured.

2 SETTING AND NOTATION

Let $f: [t_0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ be continuous and Lipschitz continuous in its second argument with constant L , that is,

$$\|f(t, u) - f(t, v)\| \leq L \|u - v\|. \quad (1)$$

By the Picard–Lindelöf theorem the initial value problem

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0, \quad (2)$$

has a unique solution y on $[t_0, T]$. For $N \in \mathbb{N}$ let $h = (T - t_0)/N$ and $t_n = t_0 + nh$. A one-step method computes approximations $y_n \approx y(t_n)$ by

$$y_{n+1} = y_n + h \Phi(t_n, y_n; h), \quad n = 0, \dots, N-1, \quad (3)$$

where Φ is the increment function. The three methods used below are

$$\Phi_{\text{Euler}} = f(t, y), \quad (4)$$

$$\Phi_{\text{Heun}} = \frac{1}{2} [f(t, y) + f(t + h, y + hf(t, y))], \quad (5)$$

$$\Phi_{\text{RK4}} = \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4), \quad (6)$$

with the stages $k_1 = f(t, y)$, $k_2 = f(t + \frac{h}{2}, y + \frac{h}{2}k_1)$, $k_3 = f(t + \frac{h}{2}, y + \frac{h}{2}k_2)$ and $k_4 = f(t + h, y + hk_3)$; see Algorithm 1 and Appendix A.

Algorithm 1: One integration with the classical Runge–Kutta method (RK4)

Input : f, t_0, y_0, T , number of steps N
Output: approximations $y_0, \dots, y_N$

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1  $h \leftarrow (T - t_0)/N$ ;
2 for  $n \leftarrow 0$  to  $N - 1$  do
3    $k_1 \leftarrow f(t_n, y_n)$ ;
4    $k_2 \leftarrow f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$ ;
5    $k_3 \leftarrow f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2)$ ;
6    $k_4 \leftarrow f(t_n + h, y_n + hk_3)$ ;
7    $y_{n+1} \leftarrow y_n + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)$ ;
8    $t_{n+1} \leftarrow t_n + h$ ;

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Definition 1 (Consistency of order p). The method (3) is *consistent of order p* if there are constants $C, h_0 > 0$ such that, for the exact solution y ,

$$\|y(t + h) - y(t) - h \Phi(t, y(t); h)\| \leq C h^{p+1} \quad (7)$$

for all $0 < h \leq h_0$ and $t \in [t_0, T - h]$.

3 GLOBAL ERROR ESTIMATE

Theorem 1. Assume that (3) is consistent of order p and that Φ is Lipschitz continuous in its second argument with constant $\Lambda > 0$, uniformly in t and $h \leq h_0$. Then the global error $e_n = y(t_n) - y_n$ satisfies

$$\|e_n\| \leq \frac{C}{\Lambda} (e^{\Lambda(t_n - t_0)} - 1) h^p \quad (0 \leq n \leq N, h \leq h_0). \quad (8)$$

Proof. Subtracting (3) from the identity $y(t_{n+1}) = y(t_n) + h\Phi(t_n, y(t_n); h) + \delta_n$, where δ_n is the local error in (7), gives

$$e_{n+1} = e_n + h [\Phi(t_n, y(t_n); h) - \Phi(t_n, y_n; h)] + \delta_n.$$

By the Lipschitz condition on Φ and by (7),

$$\|e_{n+1}\| \leq (1 + h\Lambda) \|e_n\| + C h^{p+1}.$$

Table 1. Absolute error at $t = 1$ and observed order of convergence for $y' = y$, $y(0) = 1$.

h	Euler		Heun		RK4	
	error	order	error	order	error	order
1/4	2.769×10^{-1}	–	2.343×10^{-2}	–	7.189×10^{-5}	–
1/8	1.525×10^{-1}	0.86	6.441×10^{-3}	1.86	4.984×10^{-6}	3.85
1/16	8.035×10^{-2}	0.92	1.688×10^{-3}	1.93	3.281×10^{-7}	3.93
1/32	4.129×10^{-2}	0.96	4.322×10^{-4}	1.97	2.105×10^{-8}	3.96
1/64	2.094×10^{-2}	0.98	1.093×10^{-4}	1.98	1.333×10^{-9}	3.98
1/128	1.054×10^{-2}	0.99	2.749×10^{-5}	1.99	8.384×10^{-11}	3.99
1/256	5.290×10^{-3}	0.99	6.893×10^{-6}	2.00	5.261×10^{-12}	3.99

Since $e_0 = 0$, induction on n yields

$$\|e_n\| \leq C h^{p+1} \sum_{k=0}^{n-1} (1 + h\Lambda)^k = \frac{C h^p}{\Lambda} ((1 + h\Lambda)^n - 1).$$

Finally $(1 + h\Lambda)^n \leq e^{nh\Lambda} = e^{\Lambda(t_n - t_0)}$. $\square$

Corollary 1 (Explicit Euler). *If $y \in C^2([t_0, T])$ and $M = \max_t \|y''(t)\|$, then for the explicit Euler method*

$$\|e_n\| \leq \frac{M}{2L} (e^{L(t_n - t_0)} - 1) h.$$

Proof. Here $\Phi = f$, so $\Lambda = L$ by (1). Taylor's theorem with integral remainder gives $\|y(t+h) - y(t) - hy'(t)\| \leq \frac{M}{2} h^2$, hence $p = 1$ and $C = M/2$. Apply **Theorem 1**. $\square$

Remark 1. The factor $e^{\Lambda(T-t_0)}$ reflects the worst-case amplification of perturbations over the whole interval, so the bound is pessimistic on long time horizons. The exponent p , however, matches the observed order in the experiments below.

4 NUMERICAL ILLUSTRATION

We take $d = 1$, $f(t, y) = y$, $y(0) = 1$ and $T = 1$, so that $y(t) = e^t$ and $y(1) = e$. For this problem the Euler iterates have the closed form $y_N = (1 + h)^N$, and $\Lambda = L = 1$. We integrate with $h = 2^{-k}$ for $k = 2, \dots, 8$ in double precision and report $|e - y_N|$.

Table 1 lists the errors and the observed order $\log_2(e(2h)/e(h))$. The orders tend to 1, 2 and 4, as predicted by **Theorem 1**; **Figure 1** shows the same data on logarithmic axes, where the slopes are the orders.

Cost. Accuracy has a price: Euler needs one evaluation of f per step, Heun two and RK4 four. To reach an error below 10^{-6} the three methods need 1 359 469, 673 and 13 steps, that is 1.36×10^6 , 1 346 and 52 evaluations of f . At this tolerance RK4 is therefore more than four orders of magnitude cheaper than Euler, in line with the exponent p in (8).

5 CONCLUSION

The global error of an explicit one-step method inherits the consistency order of its increment function, with a constant that grows at most exponentially in the

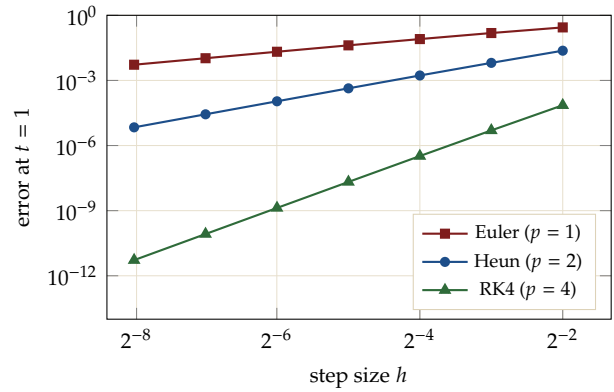


Figure 1. Error at $t = 1$ versus step size for $y' = y$, $y(0) = 1$. The slopes in the log-log plot equal the orders of the methods.

interval length. The experiments confirm the exponents 1, 2 and 4 and illustrate why high-order methods dominate when tight tolerances are required. The same estimate underlies the error control in modern adaptive solvers [4].

A BUTCHER TABLEAUX

The classical method RK4 and the method of Heun can be written as Butcher tableaux:

0					0		
$\frac{1}{2}$	$\frac{1}{2}$				1		
$\frac{1}{2}$	0	$\frac{1}{2}$			$\frac{1}{2}$	$\frac{1}{2}$	
1	0	0	1		$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{3}$
	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6}$			

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